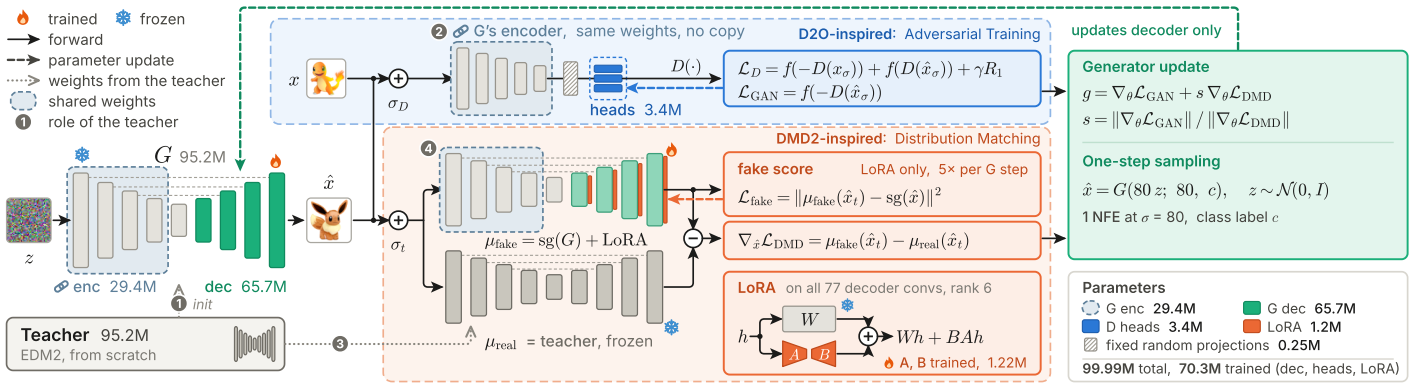


Figure 1. One-step generator G (one-NFE). One EDM2 teacher fills four roles: it initialises G (1); its frozen encoder, kept in G, is the discriminator's backbone (2); it is the real score (3); and G under stop-gradient plus a rank-6 LoRA is the fake score (4). f: softplus; sg: stop-gradient.

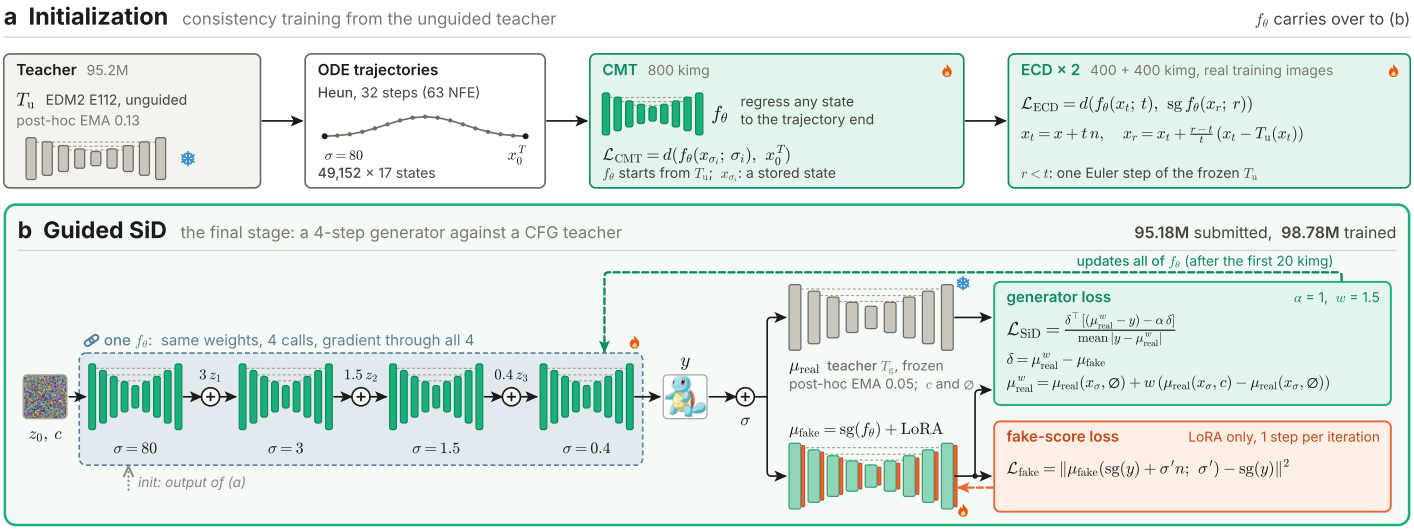


Figure 2. Four-step consistency model f_θ (few-NFE). (a) Initialisation from the unguided teacher T_u. (b) Guided SiD trains the four chained calls together against the guided teacher T_g and a fake score, f_θ under stop-gradient plus a rank-12 LoRA that only $\mathcal{L}_\mathrm{fake}$ updates; no real images are used. The chain is drawn at the training schedule; sampling uses $\sigma = (80, 5, 1.8, 0.4)$. Symbols as in Fig. 1; d: pseudo-Huber distance.

1 Overview

The task is class-conditional 64×64 Pokémon generation (151 classes) from scratch with fewer than 100M parameters, at one network evaluation (one-NFE) and at two to four (few-NFE) [1]. We trained one EDM2 teacher [2] and distilled it into two students (Table 1):

- **One-NFE:** a one-step generator G, trained with the adversarial loss of D2O [3] and the distribution-matching gradient of DMD2 [4, 5] (Fig. 1, Sec. 2).
- **Few-NFE:** a four-step consistency model f_θ [6, 7], initialised by consistency training on the teacher's trajectories and finished with guided SiD [8] (Fig. 2, Sec. 3).

Table 1. Models. Parameters at sampling time; training time and peak memory on one A100 vGPU (f_θ: one RTX 3090), after the teacher (+); teacher samples for G took another 2.4 h. Validation FID [9, 10] with fixed sampling noise, so every run gives the same value. Training logs of G and of guided SiD are in `training/d2o_one/records/` and `training/ectmodel/logs/`.

Model	NFE	Params	Time	Memory	FID
Teacher (Heun)	63	95.2M	98.1 h	17.0 GiB	20.51
One-step G	1	95.2M	+111.4 h	14.0 GiB	21.36
Four-step f_θ	4	95.2M	+9.9 h	13.6 GiB	21.32

Teacher. EDM2 preset E112 (95.2M parameters), trained for 58.7M images at batch 256 on 16,324 training images, with post-hoc EMA ($\sigma_\mathrm{rel} = 0.13$; loss in Fig. 3a). It samples with 32 Heun steps (63 NFE) [11].

Main findings.

- One teacher serves as G's initialisation, the discriminator's backbone, the real score and the base of the fake score; the one-NFE training graph holds 99.99M parameters (70.3M trained).
- The discriminator judges nearly clean images, while the DMD signal peaks at $\sigma \approx 5$. Without DMD, coarse structure degrades; without the GAN term, the mix of modes drifts.
- The conflict between the GAN and DMD gradients comes almost entirely from one scalar gain; taking that gain out of the rescaling changed no metric.
- Guided SiD through the whole four-step chain lowered every FID we track.

2 One-NFE model

2.1 Method

Generator. G starts from the teacher snapshot at 21.0M images and maps noise to an image with one call of its U-Net [12],

$$\hat{x} = G(\sigma_\mathrm{max} z;\, \sigma_\mathrm{max}, c), \qquad z \sim \mathcal{N}(0, I), \quad \sigma_\mathrm{max} = 80, \tag{1}$$

for a class c. As in D2O [3], the encoder and embeddings (29.4M) stay frozen and only the decoder (65.7M) is trained.

Discriminator. D has no backbone of its own. It reads G's frozen encoder (the same weights, not a copy) through fixed random projections and trains three class-conditional heads (3.36M) [3, 13, 14]; LADD [15] and SDXL-Lightning [16] also build their discriminators on a diffusion network. Real images x and samples $\hat{x}$ are noised as in Diffusion-GAN [17], $x_\sigma = x + \sigma_D n$ with

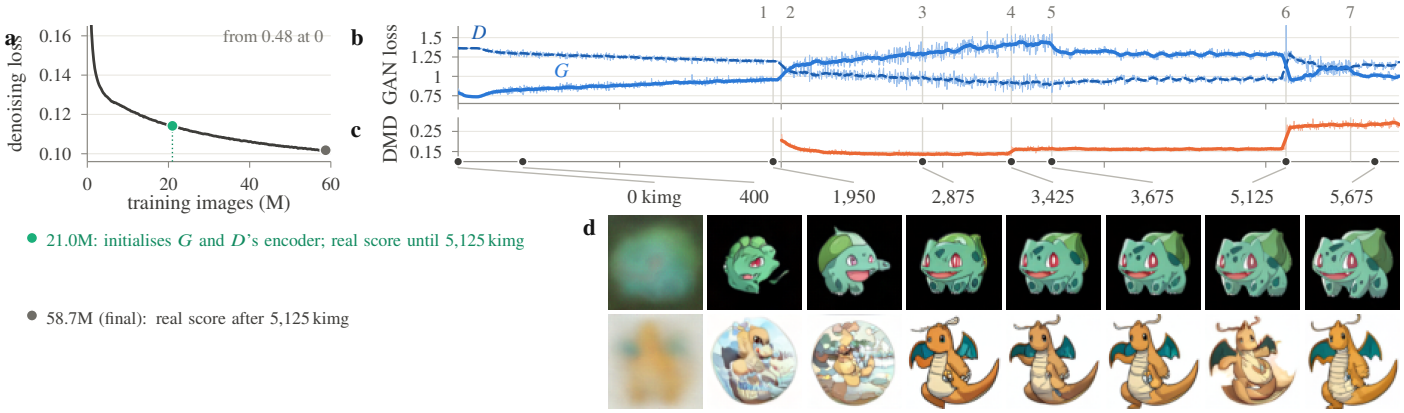


Figure 3. Training logs. (a) Teacher's denoising loss; dots: the two snapshots used later. (b) GAN losses of G (solid) and D (dashed) and (c) DMD loss, whose level shifts at stage 6 with the new real score; light: every 2 kimg, dark: mean of 25 logs. (d) EMA samples at the marked points for two latent–class pairs; at 0 kimg, G is the teacher at $\sigma = 80$. The submitted EMA ends the run, 150 kimg after the last column.

Table 2. Training stages of G. Learning rates start at 10^{-3} for G and 10^{-4} for D.

kimg	Change
0	GAN loss only
1,950	(1) fake score on; D drops one of its four heads
2,000	(2) DMD on
2,875	(3) five fake-score steps per G step instead of three
3,425	(4) real-score guidance 1.3 for $0.25 < \sigma_t \leq 4$ [26, 27], until 5,125
3,675	(5) learning rates cut to 30%
5,125	(6) teacher samples as reals, final teacher as real score, learning rates restored
5,525	(7) learning rates cut to 30% again
5,825	end of run; submitted EMA

$\ln \sigma_D \sim \mathcal{N}(-2, 1)$ clamped to $[0.02, 80]$, and the losses are non-saturating [18]:

$$\begin{aligned} \mathcal{L}_D &= f(-D(x_\sigma)) + f(D(\hat{x}_\sigma)) + \gamma R_1, \\ \mathcal{L}_{\text{GAN}} &= f(-D(\hat{x}_\sigma)), \end{aligned} \tag{2}$$

with f the softplus, lazy R1 [19, 20] and adaptive augmentation [21], whose probability never exceeded 0.013. The reals are the 17,079 training images until 5,125 kimg and 51,340 teacher samples after.

Fake score and DMD. DMD [4, 5] noises $\hat{x}$ to $\hat{x}_t = \hat{x} + \sigma_t n$ and compares two denoisers there: the teacher μ_{real} and a fake score μ_{fake} trained on G's samples,

$$\begin{aligned} \nabla_{\hat{x}} \mathcal{L}_{\text{DMD}} &= \frac{\mu_{\text{fake}}(\hat{x}_t) - \mu_{\text{real}}(\hat{x}_t)}{\text{mean}|\hat{x} - \mu_{\text{real}}(\hat{x}_t)|}, \\ \mathcal{L}_{\text{fake}} &= \lambda(\sigma_t)\, \|\mu_{\text{fake}}(\hat{x}_t) - \text{sg}(\hat{x})\|^2, \end{aligned} \tag{3}$$

with $\sigma_t \in [0.0032, 72]$ and the weight λ from EDM [11]. Instead of a separate fake-score U-Net (about 95M more trained parameters), the fake score is G under stop-gradient plus a rank-6 LoRA [22, 23] on its 77 decoder convolutions (1.22M). The DMD gradient is rescaled to the norm of the GAN gradient, a simplified form of the adaptive weight in VQGAN and CTM [24, 25]:

$$g = \nabla_\theta \mathcal{L}_{\text{GAN}} + s\, \nabla_\theta \mathcal{L}_{\text{DMD}}, \qquad s = \frac{\|\nabla_\theta \mathcal{L}_{\text{GAN}}\|}{\|\nabla_\theta \mathcal{L}_{\text{DMD}}\|}. \tag{4}$$

Training. Table 2 lists the stages (Fig. 3b, c). We submit the EMA at the end of the run (5,825 kimg), the saved checkpoint with the lowest FID to 15,100 teacher samples (10.46); its output gain of 1.10 was set on the same samples, and its latents are stratified (Sec. 3.1).

2.2 Analysis

Unless noted, we analyse the raw (non-EMA) state at 5,825 kimg.

Teacher features. A linear probe on the frozen encoder's features (null class input) classifies 3,020 held-out training images into the 151 classes with 81.4% top-1 accuracy at $\sigma = 0.135$, against 42.2% for 8×8 pixels (chance: 0.66%). The accuracy is flat over $\sigma = 0.05$ to 0.2, as reported for diffusion features [28, 29], and 0.135 is also the median of σ_D. A 1.2M LoRA is enough to turn G back into a denoiser: without it, G's EDM-weighted denoising loss on its own samples is above the teacher's at every σ (by 0.16 to 0.82), and with it, below the teacher's at all 24 levels we measured.

Where each loss acts. The GAN gradient is a fine pattern over the whole image, while the DMD gradient follows edges at small σ_t and whole parts at large σ_t (Fig. 4a). After rescaling, the GAN gradient has 2.9 times the power of the DMD gradient at the lowest spatial frequency and up to 39.7 times at high frequencies (Fig. 4b). D separates samples from reals best at $\sigma_D = 0.05$ to 0.135 (ROC-AUC 0.83) and much less at $\sigma_D = 2$ (0.68), so it judges nearly clean images. The DMD signal, the teacher's loss minus the fake score's on G's samples, peaks instead at $\sigma_t \approx 5$ (0.044, 22% of the teacher's loss), where only coarse layout and colour survive (Fig. 5a).

Ablations. From the 5,825-kimg state we trained for 60 kimg without one of the losses, each run against a control continued the same way, all with two seeds; FD is measured against a class-balanced 3,020-image training reference. We call a change real only if it exceeds twice the SD of the control's detrended snapshots at both 50 and 60 kimg; by this test the two controls differ in none of twelve metrics, while removing either loss raises FD by 2 to 4 (Fig. 5c).

- *Without DMD*, FD rose by 2.6 and 2.0, and the teacher's loss on generated images rose most at $\sigma = 2$ to 5 (+29% and +31% at $\sigma = 5$), where the DMD signal lives (Fig. 5b). FD jumped within the first 10 kimg and then stayed flat, while D's real-minus-fake logit gap kept falling (0.82 and 0.79 to 0.38; controls 0.72 and 0.76), so the lost structure explains the drop better than over-fooling.
- *Without the GAN term*, FD rose more (3.3 and 3.2), although the teacher's loss *fell* at all eight σ in both seeds (by 5% to 26%). The cluster shares of generated images moved away from those of the training images (total-variation distance 0.28 in both seeds, against 0.11 for the controls; Fig. 5d): on its own, DMD's mode-seeking reverse-KL objective [4, 30] pulls samples into the modes the teacher finds easiest.

Gradient conflict. Per sample, the two gradients on the pixels are nearly orthogonal (mean cosine −0.007 over 2,048 samples), but on the trained parameters their cosine averages −0.58 over 16 batches, the kind of conflict that gradient surgery [31] targets. Of this dot product, 99.4% comes from one learned scalar, the

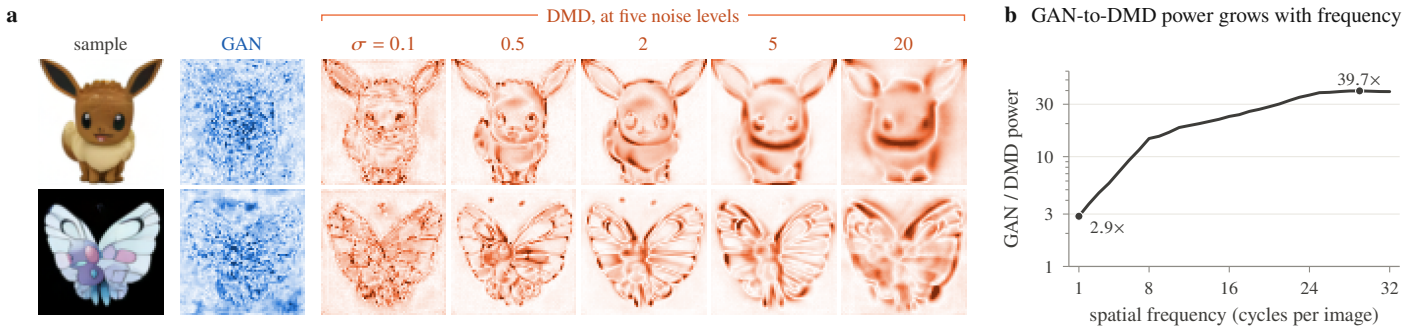


Figure 4. The two gradients in pixel space. (a) Two samples of G (chosen from 93 candidates) and the per-pixel magnitude of the GAN gradient (blue) and of the DMD gradient at five fixed σ (orange), each averaged over noise draws and scaled to its own 99.5th percentile; darker is larger. (b) Power of the GAN gradient over that of the rescaled DMD gradient, by spatial frequency.

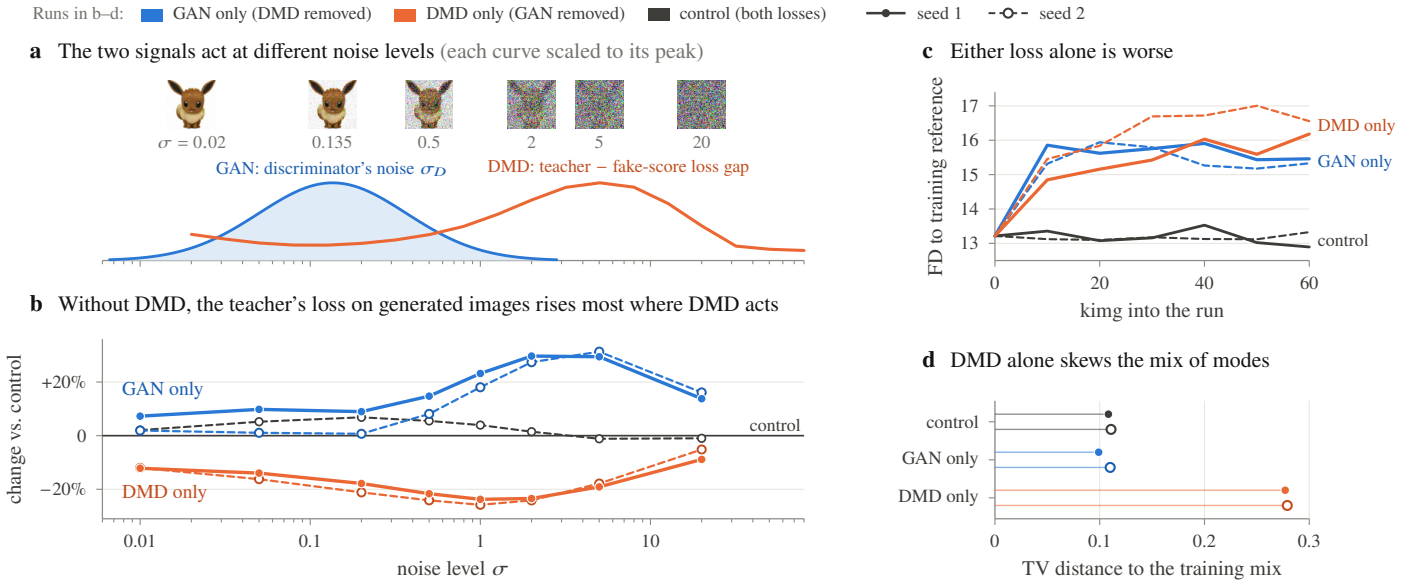


Figure 5. Where each loss acts, and what is lost without it. (a) The discriminator sees only lightly noised images (blue: its noise distribution), while the DMD signal, the teacher-minus-fake-score loss gap on G's samples, peaks at $\sigma \approx 5$ (orange); the strip shows one sample at these levels. (b)–(d) 60 kimg of training from one state with one loss removed, each against a control continued the same way, two seeds each (solid: first, dashed: second). (b) Teacher's denoising loss on generated images at 60 kimg, relative to the control of the same seed; the dark dashed line compares the two controls (run-to-run noise). (c) FD to a class-balanced 3,020-image training reference. (d) Total-variation distance between the cluster shares of generated and training images at 60 kimg (64 k-means clusters of teacher features).

decoder's output gain, which the GAN gradient pushes down (15 of 16 batches) and the DMD gradient up (16 of 16); without the 22 scalar gains the cosine is −0.14. The gain also carries 74% to 79% of each squared norm and so sets s in (4) (27.3 ± 23.3), the strength of DMD on the other 65.7M parameters. Computing s without the scalar gains made the update steadier (coefficient of variation 0.16, against 0.40 and 0.35 for the controls) but changed neither FD, the teacher's loss at any σ, nor the gain (one seed).

Student and teacher. We compared 8,154 samples of the EMA at 5,675 kimg (output gain 1.10, independent latents) with the full training split.

- FD 7.32, against 5.40 for teacher samples; precision 0.733 against 0.890 and recall 0.515 against 0.704 [32, 33].
- No copying of the 51,340 teacher samples that D sees as reals [34, 35]: in a reference set mixing 8,154 of them 1:1 with fresh teacher samples, the nearest neighbour of a generated image is one of them for 0.494 of images (95% CI 0.476 to 0.512), as expected without copying.
- The teacher's noise-to-image map is kept, though no loss asks for it: among the 16 teacher images of a class, the one closest to a student image comes from the same z in 53.9% of cases (chance: 6.25%).

3 Few-NFE model

For two to four evaluations we distilled the same teacher, in three stages, into a four-step consistency model f_θ (Fig. 2). It has the teacher's architecture (95,175,157 parameters) and shares no weights with G.

3.1 Method

Sampler. Sampling calls f_θ four times, with fresh noise before each call:

$$\hat{x}_1 = f_\theta(\bar{x} + \sigma_0 z_0;\, \sigma_0, c), \quad \hat{x}_{k+1} = f_\theta(\hat{x}_k + \sigma_k z_k;\, \sigma_k, c), \quad (5)$$

for $k = 1, 2, 3$, with $\bar{x}$ the mean training image and $(\sigma_0, \ldots, \sigma_3) = (80, 5, 1.8, 0.4)$; the output is $\hat{x}_4$. Each call predicts a clean image, and later calls change it less and at finer scales (Fig. 6a). Over 32 images, the second to fourth calls change the pixels (in $[0, 1]$) by 0.144, 0.086 and 0.037 RMS, and 30%, 18% and 10% of these changes survive 8×8 block averaging.

Stratified noise. `evaluate.py` requests the 20 images of a class together, and the FID of a finite sample is biased upward by the sample's own scatter [36, 37], so both samplers spread the 20 images at low frequency. The 8×8 block means of each z_k (for G, of z), times 8, are i.i.d. standard normal; we redraw their 192 per-channel DCT coefficients over the 20 images by Latin hypercube sampling [38] and keep the high-frequency residual. Each image's noise stays $\mathcal{N}(0, I)$, and a fixed seed (the best of eight on HO) makes every run identical. The gain on the training-side FIDs HO and TR (Sec. 3.2) is small: for G, HO fell by 0.18 and TR by 0.19 on average, and both fell for each of the eight seeds; for f_θ, HO fell by 0.04 in all, and TR by 0.18 when only z_0 was stratified.

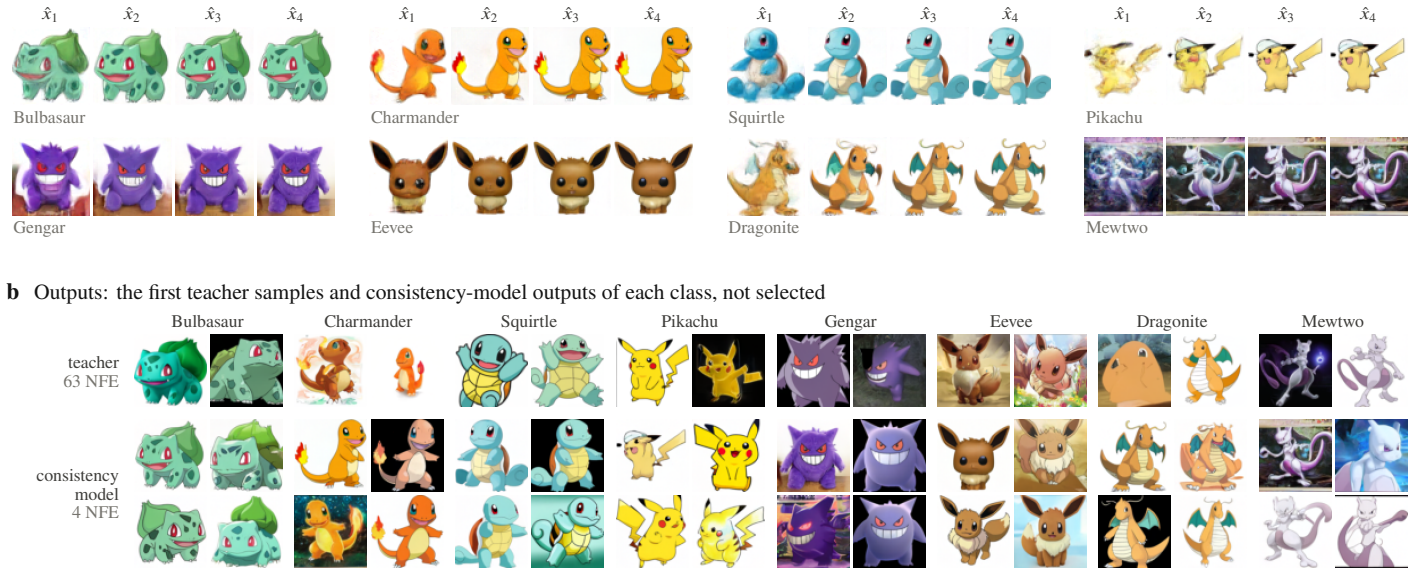


Figure 6. Samples of the four-step model. (a) The prediction $\hat{x}_k$ of each call for the first image of eight classes (computed on CPU). (b) The first two teacher samples (32 Heun steps) and the first four outputs of f_θ (one `evaluate.py` run) for each class, not selected.

CMT. A denoiser predicts a conditional mean, which blurs when chained. Consistency mid-training [39] therefore first regresses 17 states x_{σ_i} of each of 49,152 trajectories of the teacher's probability-flow ODE [40] (32 Heun steps) onto its end x_0^T, $\mathcal{L}_{\text{CMT}} = d(f_\theta(x_{\sigma_i};\sigma_i), x_0^T)$, where knowledge distillation [41] uses only the first state. The distance d is pseudo-Huber [42] in place of LPIPS [43]. We use the snapshot at 800 kimg (batch 128), taken before the set grew to 65,536 for a continuation we did not use.

ECD. Regression on fixed trajectories does not make every noise level consistent; consistency distillation [6, 44] does. A training image x is noised to $x_t = x + t\,n$ at a log-normal level t, one Euler step of the teacher's ODE gives a slightly lower level r, and

$$\mathcal{L}_{\text{ECD}} = d\big(f_\theta(x_t;t),\ \text{sg}\, f_\theta(x_r;r)\big), \qquad x_r = x_t + \tfrac{r-t}{t}\big(x_t - T_u(x_t)\big). \tag{6}$$

We ran 400 kimg at a learning rate of 10^{-4} and 400 kimg at $2{\times}10^{-5}$ (batch 128).

Guided SiD. ECD trains one call at a time, so the four chained calls are never optimised as a whole. The last stage back-propagates through all four [45] and matches the distribution of their output y by Score identity Distillation [8], which needs no real images. Both denoisers see $x_\sigma = y + \sigma n$, and with $\alpha = 1$ the generator loss is

$$\mathcal{L}_{\text{SiD}} = \frac{\delta^\top\big[(\mu^w_{\text{real}} - y) - \alpha\,\delta\big]}{\text{mean}|y - \mu^w_{\text{real}}|}, \qquad \delta = \mu^w_{\text{real}} - \mu_{\text{fake}}, \tag{7}$$

with the denominator, a per-image weight, held fixed. The real score μ^w_{real} is the teacher (a shorter EMA, $\sigma_{\text{rel}} = 0.05$) with classifier-free guidance $w = 1.5$ [26, 46], so guidance adds no evaluations at sampling time. As for G, the fake score is f_θ under stop-gradient plus a LoRA (rank 12, 3.61M), trained to denoise y. Training then holds 98,781,577 parameters instead of about 190M with a separate fake network, at no measurable cost (HO 18.56, against 18.58 with a full copy). SiD ran for 70 kimg at the schedule $\sigma = (80, 3, 1.5, 0.4)$ with batch 64 and learning rates $3{\times}10^{-6}$ for f_θ and 10^{-4} for the fake score, which trained alone for the first 20 kimg (Fig. 7b). On one RTX 3090 the trajectories took 2.8 h, CMT 1.8 h, ECD 3.2 h and SiD 2.1 h.

3.2 Results

We track two training-side FIDs: HO, against 2,265 training images, one from each of 15 near-duplicate groups per class (18% of training images have a near-copy within a pixel RMSE of 0.01), and TR, against 10,000 teacher samples. The SiD snapshot (70 kimg, lowest mean HO over four noise seeds; Fig. 7b), the sampling schedule and the noise seed were chosen on HO. HO fell at every stage (Fig. 7a), and guided SiD, the only stage that trains the four calls together, lowered all three measures (Table 3). Over eight noise seeds the four-step model averages a validation FID of 21.39 and the one-step generator 21.62, after 9.9 h of training against 111 h (samples in Fig. 6b).

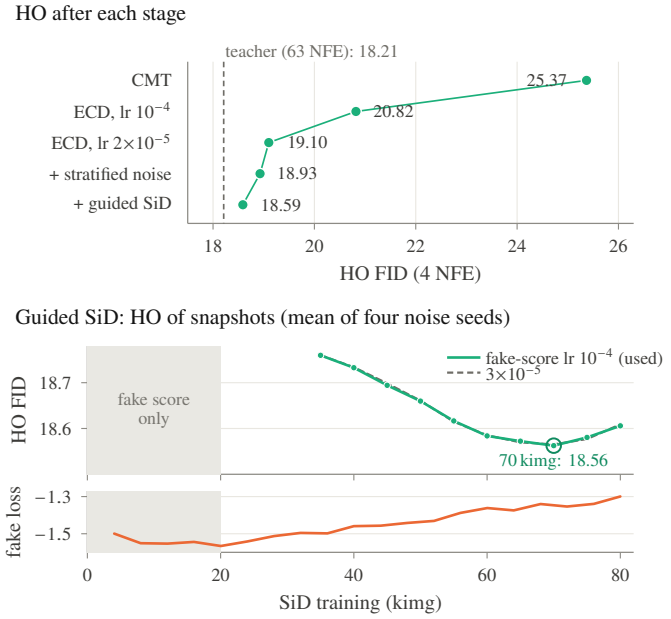


Figure 7. Training the four-step model. (a) HO after each stage; CMT and ECD sampled at $\sigma = (80, 4, 1.5, 0.5)$, the last two at their final schedules with stratified noise (means over eight noise seeds). (b) Guided SiD: HO of snapshots for fake-score learning rates 10^{-4} (used) and $3{\times}10^{-5}$, and the fake-score loss (uncertainty-weighted [47, 2]); shaded: the fake score trains alone.

Table 3. Effect of guided SiD (stratified noise): mean (range) over eight noise seeds, and the submitted seed. Schedules: $(80, 3, 1.5, 0.4)$ for ECD, $(80, 5, 1.8, 0.4)$ after SiD.

	HO	TR	Val FID
ECD	18.93	9.78	22.03 (21.51–22.40)
+ guided SiD	18.59	9.53	21.39 (21.07–21.68)
submitted seed	18.38	9.53	21.32

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